

# Mathematical Physics I (Fall 2025): Homework #4

Due Nov. 7, 2025 (Fri, 23:00pm)

[0.5 pt each, total 5 pts, turn in as a single pdf file to eTL]

- By turning in your homework, you acknowledge that you have not received any unpermitted aid, nor have compromised your academic integrity during its preparation. (Remember the SNU College of Natural Sciences Honor Code!)
- Textbook problem numbering convention (Boas, 3rd ed.): [Section].[Problem No.]. Many problems in this assignment are from the list of suggested problems announced in the class.
- Exhibit all intermediate steps to receive full credits. Only handwritten answers are accepted except for numerical problems or verifications — for which you print out and turn in not just the end results (e.g., plots) but also the source codes in your favorite tools such as MATLAB or MATHEMATICA.

1.-6. Boas Chapter 7, Problems 5.9, 6.12 (for Problem 5.7), 7.7, 8.15(a), 11.8, 12.15

(Note: For Problems 5.9, you are asked to tackle the problem in two ways — first directly find the Fourier coefficients, then verify your result using the answer given in Problem 5.7. For Problems 6.12 and 7.7, you will first have to explicitly work out Problem 5.7. Then for Problem 6.12, you may consider writing a simple script in MATLAB or in `python` to automatically plot several partial sums for different values of  $n$ 's. However, creating such an automated script is not required to receive full credit. For Problem 11.8, you are asked to first work out Problem 9.10. For Problem 12.15, you are asked to first work out Problem 12.9.)

7.-8. Boas Chapter 8, Problems 2.6, 3.11

(Note: For Problem 2.6, read the instruction in the textbook carefully; that is, you have been asked to plot a slope field and the solution curve with a computer. To learn more about the “slope field”, you may review Boas Chapter 8, Section 1. For Problem 3.11, again, read the instruction in the textbook carefully.)

9. Here we study the Fourier series expansion of an example of an impulse-like function, and examine its behavior.

(a) Consider an impulse-like function of width  $\frac{1}{N}$ , given in the interval  $-\pi < x < \pi$ :

$$\delta_N(x) = \begin{cases} N, & \text{if } |x| < \frac{1}{2N}, \\ 0, & \text{if } |x| > \frac{1}{2N}. \end{cases}$$

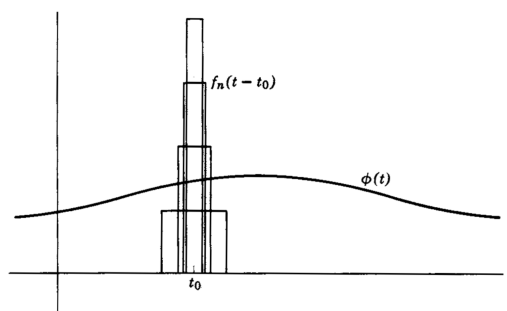
Following the worked Example in Boas Chapter 7, Section 9 (or Problems 9.17-9.22 in Boas Chapter 7), expand  $\delta_N(x)$  in a Fourier cosine series.

(b) What does your Fourier series from (a) converge to in the limit as  $N \rightarrow \infty$ ?

(c) Show that for any function  $f(x)$  that is finite in the interval  $[-\pi, \pi]$  and continuous at  $x = 0$ ,

$$\lim_{N \rightarrow \infty} \int_{-\pi}^{\pi} f(x) [\text{Fourier series expansion of } \delta_N(x)] dx = f(0),$$

which resembles an important property of what we later define as the Dirac delta function,  $\delta(x)$ .



[Adapted from Boas, Chapter 8.11]

(Note: You may find it useful to preview Eq.(11.6) in Boas Chapter 8, along with the accompanying discussion, which illustrates this property of the Dirac delta function.)

10. Let us use the Fourier transform to appreciate the meaning of the Heisenberg uncertainty principle in quantum mechanics.

(a) Imagine an infinite wave train  $\sin \omega_0 t$  clipped by shutters to maintain only  $N$  cycles of the original waveform:

$$f(t) = \begin{cases} \sin \omega_0 t, & \text{if } |\omega_0 t| < N\pi, \\ 0, & \text{if } |\omega_0 t| > N\pi. \end{cases}$$

Find the amplitude function of the Fourier (exponential) transform,  $g(\omega)$ . Since the prefactor may depend on the exact definition of the transform, do not worry too much about it.

(b) Find the amplitude function of the Fourier sine transform,  $g_s(\omega)$ . Again, since the prefactor may depend on the exact definition of the transform, do not worry too much about it.

(c) Show that, in the special case of  $N = \frac{1}{2}$  and  $\omega_0 = 1$ ,  $g_s(\omega)$  becomes

$$g_{s,1}(\omega) = \sqrt{\frac{2}{\pi}} \frac{\omega \cos\left(\frac{\omega\pi}{2}\right)}{1 - \omega^2}.$$

[Problem 10 continues in the next page.]

(d) Now consider the limit of  $\omega_0 \gg 1$  and  $\omega \approx \omega_0$ . Show that  $g_s(\omega)$  is approximately equal to

$$g_{s,2}(\omega) = \frac{1}{\sqrt{2\pi}} \frac{\sin[(\omega_0 - \omega) \frac{N\pi}{\omega_0}]}{\omega_0 - \omega}.$$

Sketch or computer plot this function for  $N = 1, 3, 5, 10$  and an arbitrary  $\omega_0$ . Notice that for  $N \gg 1$ ,  $g_{s,2}(\omega)$  may be interpreted as proportional to what we later define as the Dirac delta function,  $\delta(\omega - \omega_0)$ , in Boas Chapter 8, Section 11.

(Note: You do not need to provide a mathematically rigorous proof that  $g_{s,2}(\omega)$  indeed becomes proportional to  $\delta(\omega - \omega_0)$  as  $N \rightarrow \infty$ . For now, observe the shape of  $g_{s,2}(\omega)$  as you vary  $N$ , and based on your observation simply argue that  $\lim_{\omega \rightarrow \omega_0} g_{s,2}(\omega) \rightarrow \infty$  as  $N \rightarrow \infty$ .)

(e) Show that the first zeros of  $g_{s,2}(\omega)$  from  $\omega = \omega_0$  are at  $\omega = \omega_0 \pm \Delta\omega = \omega_0 \pm \frac{\omega_0}{N}$ . Justify that  $\Delta\omega = \frac{\omega_0}{N}$  could be a good measure of the spread (or uncertainty) in frequency of our clipped wave train. Then, establish the inverse relationship between the wave train's pulse length ( $N\pi$ ) and the frequency spread ( $\Delta\omega$ ). Finally, using the relationship along with the assumed *wave nature of matter*, explain the uncertainty principle of quantum mechanics.

(Note: You may want to briefly review the textbooks in quantum mechanics such as Griffiths & Schroeter. Note that the inverse relationship found here is a fundamental property of the finite wave train, and has little to do with any additional *ad hoc* postulates in quantum mechanics. If needed, you must reference your sources appropriately with a proper citation convention, but your answer must still be your own work in your own words. To access the electronic resources — e.g., academic journals — off-campus via SNU library's proxy service, see <http://library.snu.ac.kr/using/proxy>.)